

requires talent, judgment, intuition and other personal qualities. They admit that no rules of procedure can be laid down, the automatic following of which will insure success. Hence the widespread tendency in Wall Street circles towards a composite or eclectic approach, in which a very thorough study of the market's performance is projected against the general economic background, and the whole is subjected to the appraisal of experienced judgment.

**The Second Type of Mechanical Forecasting.** Before considering the significance of this injection of the judgment factor, let us pass on to the other type of mechanical forecasting, which is based upon factors outside of the market itself. As far as the general market is concerned, the usual procedure is to construct indices representing various economic factors, e.g., money rates, carloadings, steel production, and to deduce impending changes in the market from an observation of a recent change in these indices.<sup>3</sup> One of the earliest methods of the kind, and a very simple one, was based upon the percentage of blast furnaces in operation.

This theory was developed by Col. Leonard P. Ayres of the Cleveland Trust Company and ran to the effect that security prices usually reached a bottom when blast furnaces in operation declined through 60% of the total and that conversely they usually reached a top when blast furnaces in operation passed through the 60% mark on the upswing in use thereof.<sup>4</sup> A companion theory of Colonel Ayres was that the high point in bond prices is reached about 14 months subsequent to the low point in pig-iron production and that the peak in stock prices is reached about two years following the low point for pig-iron production.<sup>5</sup>

This simple method is representative of all mechanical forecasting systems, in that (1) it sounds vaguely *plausible* on the basis of *a priori*

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<sup>3</sup> These indices may also be plotted on charts, in which case the forecasting takes on the aspect of chart reading. *Examples:* The A, B, and C lines of the *Harvard Economic Service* which were published in weekly letters from Jan. 3, 1922, to Dec. 26, 1931 (since continued through 1939 at less frequent intervals in *The Review of Economic Statistics*); also the single composite Index Line in the "Investment Timing Service" offered by Independence Fund of North America, Inc., in 1939.

<sup>4</sup> See *Bulletin of the Cleveland Trust Company*, July 15, 1924, cited by David F. Jordan, in *Practical Business Forecasting*, p. 203n, New York, 1927.

<sup>5</sup> See *Business Recovery Following Depression*, a pamphlet published by the Cleveland Trust Company in 1922. The conclusions of Colonel Ayres are summarized on p. 31 of the pamphlet.